Received: 9 September 2024 | Revised: 29 September 2024 | Accepted: 9 October 2024

DOI: 10.1002/ccd.31272

ORIGINAL ARTICLE - CLINICAL SCIENCE

WILEY

Percutaneous endovascular management of Angio-Seal related vascular occlusion

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Abstract

Background: The use of vascular closure devices (VCDs) to achieve quick and safe hemostasis after femoral arterial access is widely accepted. Major complications include bleeding and occlusion of the femoral artery due to device failure, which often necessitates vascular intervention. This manuscript details our peripheral percutaneous endovascular interventional (PEI) approach for the management of femoral artery occlusion resulting from Angio-Seal (Terumo, Somerset, New Jersey, USA) VCD deployment.

Methods: Consecutive patients who developed occlusive complications after Angio-Seal deployment underwent PEI to overcome specific complications. Patients' clinical and procedural characteristics, along with their short- and long-term follow-up data, were analyzed.

Results: The study cohort included 40 patients who experienced Angio-Seal occlusive complications between July 2013 and September 2023. The mean age of the patients was 74 ± 10 years and 55% were female. All the patients were treated with PEI, with an overall procedural success rate of 100%. The primary approach for PEI was directional atherectomy, which was used in 35 cases (88%), followed by balloon, while a cutting balloon was used in 5 patients (13%). Stenting served as the definitive therapy in only 7 patients (18%). No procedural complications or conversions to surgery were observed. During a median follow-up of 244 (IQR = 100–707) days, none of the patients required re-intervention related to Angio-Seal occlusion and salvage intervention.

Conclusion: In the management of Angi-Seal VCD-related femoral artery occlusion, the adjunctive use of directional atherectomy followed by balloon angioplasty was effective and safe, allowing re-establishment of flow with excellent long-term outcomes.

Abbreviations: BAV, balloon aortic valvuloplasty; CFA, common femoral artery; CVD, cardiovascular disease; DSA, digital subtraction angiography; EIA, external iliac artery; IEA, inferior epigastric artery; IVUS, intravascular ultrasound; PCI, percutaneous coronary intervention; PFA, profunda femoris artery; POBA, plain old balloon angioplasty; PPI, permanent pacemaker implantation; SFA, superficial femoral artery; VCD, vascular closure devices.

KEYWORDS

Angio-Seal, atherectomy, femoral artery occlusion, percutaneous endovascular intervention, vascular closure device

1 | INTRODUCTION

The use of vascular closure devices (VCDs) is widely accepted for achieving fast and safe hemostasis following percutaneous procedures across various conditions.[1,2] Compared to manual compression, VCDs enable quicker discharge and ambulation due to improved hemostasis rates.[3] However, VCD use can lead to adverse vascular events such as stenosis, hematoma, bleeding, arteriovenous fistula, pseudoaneurysm, thrombosis, and infection.[4–6] The most serious complications include bleeding and total femoral artery occlusion, resulting in acute limb ischemia with an incidence of 0.3–0.5%, presenting acutely, sub-acutely, or chronically post-closure.[5,7] Despite low complication rates, each event is clinically significant and often requires reintervention, either via endovascular or surgical approaches.

The most commonly used VCD is the Angio-Seal device (Terumo Interventional Systems, Somerset, New Jersey, USA).[8] Initial clinical trials of this device showed ease of use, early ambulation, lower complication rates, improved patient satisfaction, and higher hemostasis success rates compared to manual compression and other VCDs.[9,10] Despite these advantages, Angio-Seal can still lead to access-site complications mainly bleeding which usually respond to prolonged manual compression and total occlusion of the femoral artery. Therefore, it is imperative to characterize these complications to prevent potential harm to patients and to improve outcomes. Herein, we outline our percutaneous peripheral endovascular interventional (PEI) approach for the management of femoral artery occlusion resulting from Angio-Seal VCD deployment.

2 | METHODS

2.1 | Study design

In this retrospective investigation conducted at a single high-volume center, data were collected from successive patients who underwent catheterization procedures using femoral arterial access complicated by VCD-related occlusion and were managed using a percutaneous endovascular approach. We included coronary, peripheral, or structural procedures involving femoral artery access that utilized Angio-Seal of both 6 and 8 French sizes for primary closure. We referred to the procedure where the Angio-Seal device was deployed and caused the obstruction as the "index procedure" and the procedure performed to resolve the occlusive complication as the "PEI procedure." Patients who underwent primary closure with additional VCD or large-bore access were excluded. Individuals

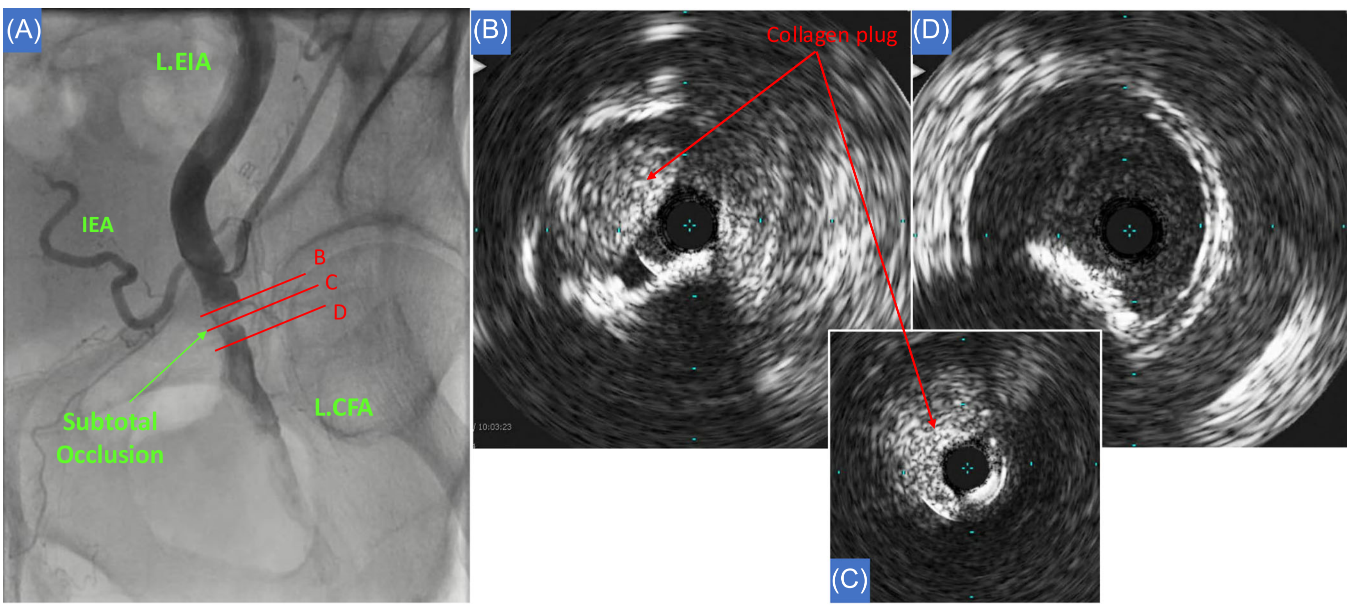

FIGURE 1 Selective iliofemoral angiography and IVUS. (A) Selective Ilio-femoral angiography performed through contralateral access demonstrating subtotal CFA occlusion. (B) IVUS of proximal subtotal occlusion demonstrating atherosclerotic disease, calcified nodule (*) and collagen plug. (C) The collagen plug is almost completely obstructing the lumen at the narrowest site. (D) Patent vessel distal to the plug location. [Color figure can be viewed at wileyonlinelibrary.com]

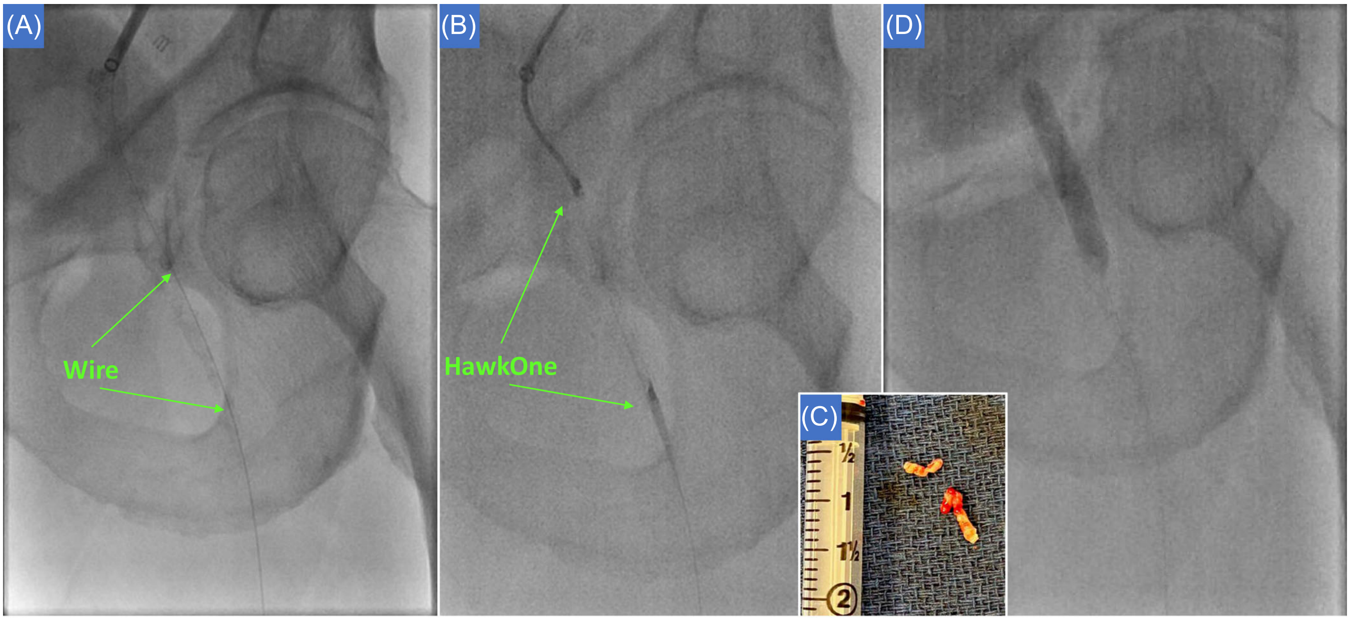

FIGURE 2 Percutaneous endovascular intervention. (A) The lesion is crossed with guidewire antegradely. (B) Directional atherectomy with HawkOne catheter was used to debulk the lesion. (C) Plug particles extracted from the HawkOne collection chamber. (D) The lesion is inflated with 1:1 balloon to vessel balloon. [Color figure can be viewed at wileyonlinelibrary.com]

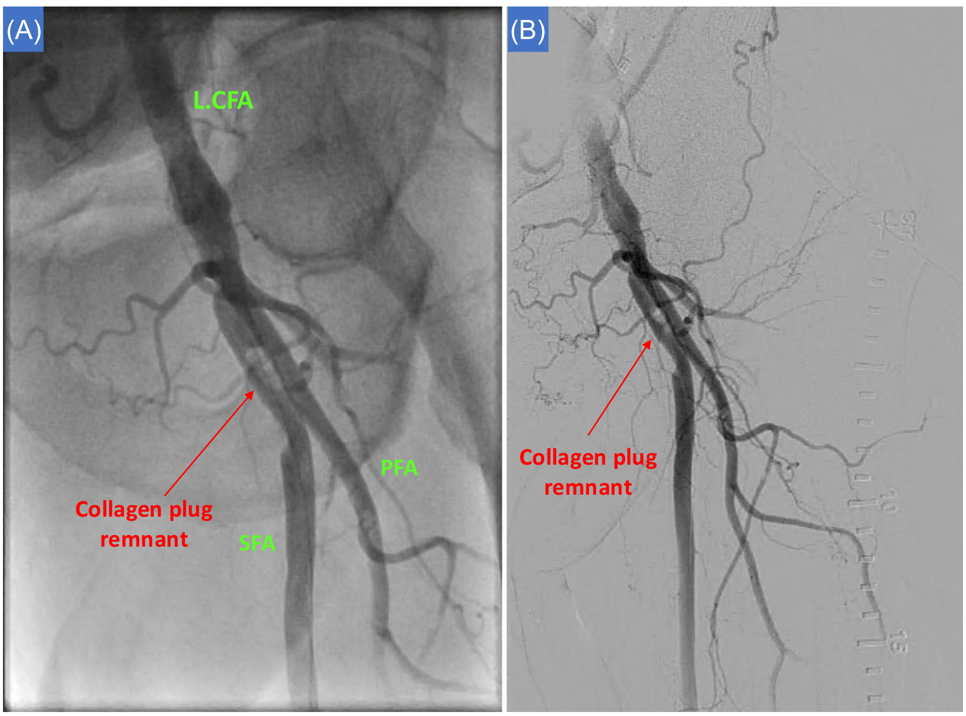

FIGURE 3 Following percutaneous endovascular intervention. (A) Post PEI cine. (B) DSA angiography showing patent lumen at CFA, adequate flow and embolized remnant of collagen plug. [Color figure can be viewed at wileyonlinelibrary.com]

who had a procedural access site other than the femoral artery and/or developed nonocclusive complications were also excluded. Patients who underwent subsequent PEI revascularization were included in this study.

Retrospective data were extracted from clinical documentation, medical chart review, procedural logs, imaging databases, and clinic visits. Baseline clinical characteristics, index and PEI procedural characteristics, and short and long-term follow-up findings of

TABLE 1 Patient demographic and clinical characteristics.

	Mean (± SD)
Age	74 ± 10 years
Female sex	22 (55)
Comorbidities	**Prevalence (*n*, %)**
Hypertension	36 (90)
Diabetes mellitus	19 (48)
Hyperlipidemia	34 (85)
Current or former smoker	19 (48)
Chronic kidney disease (GFR <60)	15 (38)
Cardiovascular disease history	**Prevalence (*n*, %)**
LVEF < 40%	10 (26)
Prior PCI	13 (33)
Prior CABG	13 (40)
Prior stroke or TIA	4 (10)
Prior MI	14 (35)
PAD	16 (40)
Carotid artery disease	8 (20)
Baseline labs	**Mean (± SD)**
Creatinine	1.2 ± 0.81 mg/dL
Hemoglobin	11 ± 2.14 g/dL
Medications	**Prevalence (*n*, %)**
Aspirin	26 (68)
Dual-antiplatelet therapy	15 (39)
Anticoagulation	18 (47)

Abbreviations: CABG, coronary artery bypass grafting; GFR, glomerular filtration rate; LVEF, left ventricular ejection fraction; MI, myocardial infarction; PAD, peripheral artery disease; PCI, percutaneous coronary intervention; SD, standard deviation; TIA, transient ischemic attack.

the patients were collected. Regarding the timeline of occlusion diagnosis, acute diagnosis was defined as within the same hospital stay, events occurring within 2 weeks post-discharge were designated as subacute, and chronic diagnosis was defined as 2 weeks post-discharge. Angiograms and catheterization reports were examined by two interventional cardiologists (NB and DH).

Categorical variables are presented as numbers and percentages, and continuous variables are presented as median values and interquartile ranges or mean (standard deviation) values. Procedural complications and outcomes were reviewed and compared. Statistical analysis was performed using SPSS version 29 software (IBM, Chicago, IL, USA). This study was conducted in accordance with the Declaration of Helsinki and was approved by the IRB and ethics committees of MedStar Washington Hospital Center (Washington, DC, USA).

2.2 | Percutaneous endovascular intervention description

If a patient developed symptoms after the use of a VCD, duplex ultrasound was used to assess the presence of significant stenosis, followed by angiography if indicated. In cases of acute limb ischemia, angiography was performed, with or without duplex ultrasonography. All PEI procedures were performed using ultrasound guidance, along with a micropuncture kit and angiography to confirm the correct height of the access point and characteristics of the vessel. Selective iliofemoral artery angiography was typically performed via crossover from contralateral femoral artery vascular access. If necessary, diagnostic angiography was complemented with intravascular ultrasound (IVUS) (Figure 1). PEI was initiated by traversing the lesion with a guidewire using an anterograde approach and wire escalation, whenever necessary. A retrograde approach was used if the anterograde approach was unsuccessful. Once the lesion was crossed, an embolic protection device, SpiderFX (Medtronic, Minneapolis, MN, USA) or Emboshield NAV 6 (Abbott Vascular, Abbott Park, IL, USA), was deployed distally in the superficial femoral artery). Subsequently, directional atherectomy with HawkOne or TurboHawk (Medtronic, Minneapolis, MN, USA) was performed to debulkthe obstructing lesion. Additional atherectomy runs or balloon inflation were conducted based on the residual lesion and blood flow (Figures 2 and 3). Stent implantation was considered optional, and the choice of embolic protection and atherectomy device was left to the operator's discretion. The standard follow-up timetable for treated patients included appointments at 1 month and 1 year. Clinical and physical assessments were performed during these visits.

3 | RESULTS

Between July 2010 and September 2023, 12,400 patients underwent catheterization procedures using femoral arterial access and were closed using Angio-Seal VCDs. Of these patients, 40 (0.32%) were found to have occlusive Angio-Seal complications and underwent PEI. The patient demographics and clinical characteristics are shown in Table 1. The mean age of the patients was 74 ± 10 years and 55% were female. In this cohort, 90% of the patients had hypertension, 48% had diabetes mellitus, 85% had hyperlipidemia, and 38% had chronic kidney disease. Of the 40 patients, all but 5 had a history of cardiovascular disease (CVD, 88%). Of the cohort, 33% had previous percutaneous coronary intervention (PCI) and 40% had peripheral vascular disease. Regarding medications, 68% of patients were on aspirin, 39% were on dual-antiplatelet therapy, and 47% were on anticoagulation (Figure 4).

In this cohort, the index procedure was coronary angiography, PCI, PEI, or balloon aortic valvuloplasty which comprised 55%, 22%, 13%, and 10%, respectively (Central Illustration). The sheath sizes used were 5 or 6 F in 60% of cases, 7 or 8 F in 30% of cases, and 10 F in 10% of cases. Vascular closure was achieved using Angio-Seal in all

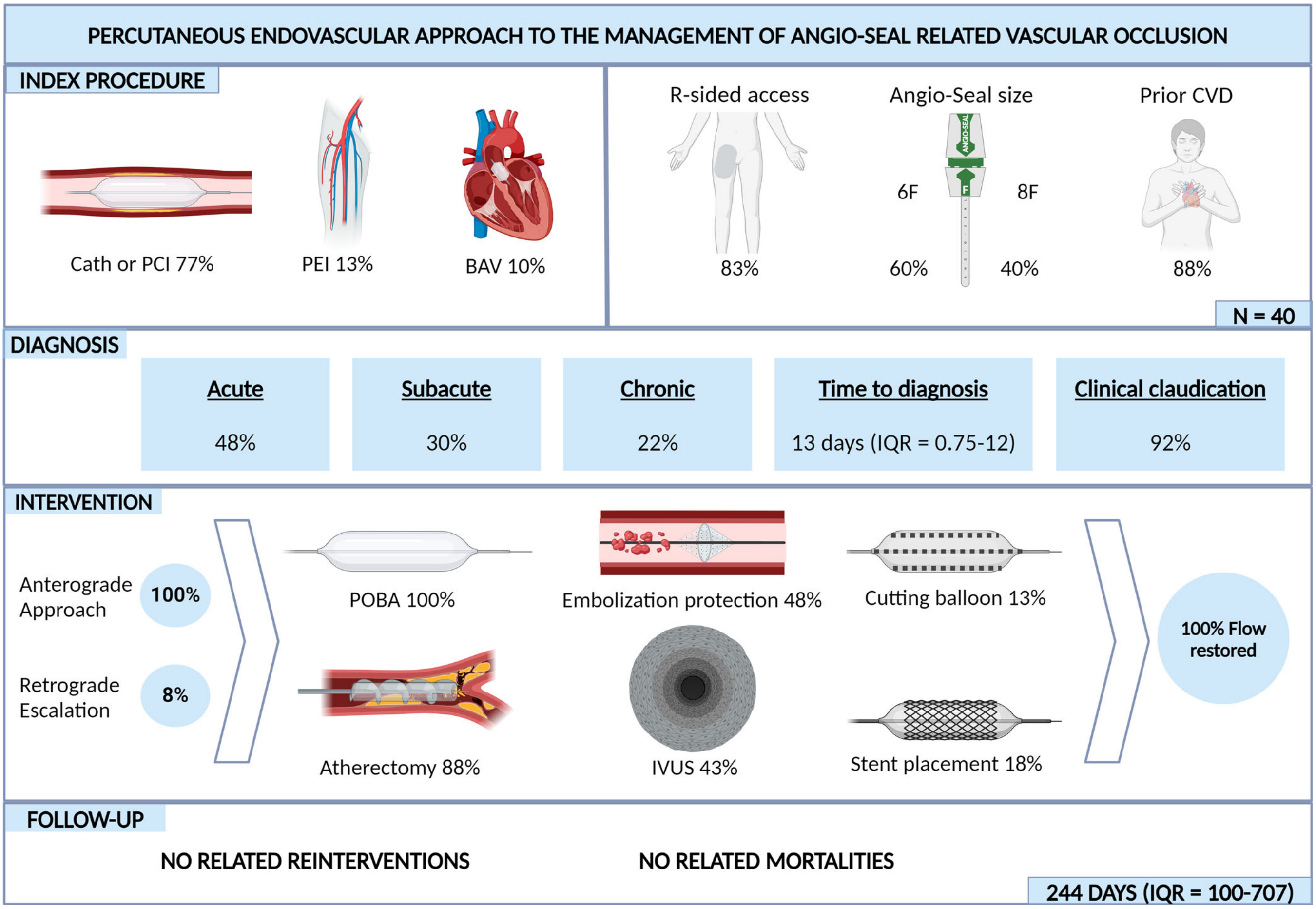

FIGURE 4 Central illustration: Percutaneous endovascular approach to the management of Angio-Seal related vascular occlusion. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 2 Index procedure with Angio-Seal closure.

Intervention type	Prevalence (*n*, %)
Coronary angiography	22 (55)
PCI	9 (22)
PEI	5 (13)
Balloon aortic valvuloplasty	4 (10)
Sheath size (F)	**Prevalence (*n*, %)**
5	11 (28)
6	13 (33)
7	2 (5)
8	10 (25)
10	4 (10)
Vascular access	**Prevalence (*n*, %)**
Access side (right)	33 (83)
Angio-Seal 6 F	24 (60)
Angio-Seal 8 F	16 (40)
Contralateral access required	3 (8)

Abbreviations: F, French; PCI, percutaneous coronary intervention; PEI, peripheral endovascular intervention.

cases; 6 F Angio-Seal was used in 60% of cases and 8 F in 40% of cases. The complete characteristics of the index procedure are presented in Table 2.

All participants had occlusive vascular complications following the use of Angio-Seal VCD. Occlusions were acute in 19 cases (48%), subacute in 12 cases (30%), and chronic in 9 cases (22%). Claudication symptoms were observed in 36 (92%) patients, and 26 (65%) presented with acute limb ischemia. Angiography revealed that 35 patients (88%) had common femoral artery occlusion, while 5 patients (12%) had proximal superficial femoral artery (SFA) occlusion (Tables 3 and 4).

All cases were initially approached anterogradely through the contralateral iliofemoral artery, and 3 (8%) patients required an additional retrograde approach through the ipsilateral SFA to successfully advance the guidewire across the lesion. The primary approach for PEI was debulking atherectomy, which was applied in 35 (88%) cases, mostly using HawkOne or TurboHawk directional atherectomy (75%). Balloon angioplasty was performed in all cases. Definitive therapy with cutting balloon angioplasty was used in 5 (13%) patients, and stenting was performed in 7 (18%) patients. Protection against embolization was used in 48% of the participants, and intravascular ultrasound (IVUS) was used to better understand

TABLE 3 Angio-Seal related occlusion and revascularization characteristics.

Time to diagnosis	Prevalence (*n*, %)
Acute	19 (48)
Subacute	12 (30)
Chronic	9 (22)
Median days to diagnosis	13 (IQR 0.75–12)
Clinical characteristics	**Prevalence (*n*, %)**
Rutherford stage	
1	1 (3)
2	3 (8)
3	14 (35)
4	19 (48)
5	2 (5)
Claudication present	36 (92)
Acute limb ischemia	26 (65)
Angiography results	**Prevalence (*n*, %)**
Complete (total) vessel occlusion	23 (58)
Subtotal vessel occlusion	12 (30)
Common femoral artery occlusion	35 (88)
Superficial femoral artery occlusion	5 (13)
TIMI flow score	
0	12 (30)
1	9 (23)
2	12 (30)
3	7 (17)
Revascularization characteristics	**Prevalence (*n*, %)**
Anterograde/contralateral approach	40 (100)
Escalation to retrograde approach	3 (8)
Balloon angioplasty	40 (100)
Atherectomy performed	35 (88)
HawkONE or TurboHawk	30 (75)
JetStream	5 (13)
Cutting balloon only	5 (13)
Stent placement	7 (18)
Embolization protection	19 (48)
Intravascular ultrasound	17 (43)
Residual stenosis ≥30%	2 (7)
Flow restored	40 (100)

Abbreviations: SD, Standard deviation; TIMI, Thrombolysis in Myocardial Infarction.

TABLE 4 Follow-up data after Angio-Seal occlusion reintervention.

	***n* = 32**
Median follow-up (days)	244 (IQR = 100–707)
	Prevalence (*n*, %)
Claudication unrelieved by reintervention	3 (10)
Occlusion site reinterventions	0
Unrelated mortalities	3 (9)

the occluding mechanism or verify the guidewire location in 17 patients (43%). PEI achieved successful revascularization in all cases (100%). No severe procedural complications or conversions to surgery were reported.

During a median follow-up of 244 (IQR 100–707) days, none of the patients required re-intervention related to the Angio-Seal occlusion or the reintervention procedure. Three deaths, unrelated to vascular complications, were observed.

4 | DISCUSSION

In this manuscript, we describe our experience with percutaneous treatment of occlusive complications with the Angio-Seal VCD at a single high-volume center. Our study's main findings were as follows: (1) all patients underwent successful percutaneous endovascular intervention using the described technique, with restoration of brisk flow to the distal vessels; (2) no conversion to open surgery was necessary; and (3) midterm follow-up showed no need for further intervention. To our knowledge, this is the largest consecutive series of patients with a percutaneous bailout for acute limb ischemia following Angio-Seal related vascular occlusion.

The Angio-Seal VCD is effective in achieving vascular hemostasis but carries a rare risk of serious vascular complications, even in non-atherosclerotic femoral arteries, due to its intravascular and extravascular components. During deployment, a polymer anchor is placed intravascularly, with polylactide/polyglycolide copolymer and collagen foam designed to reduce short-term thrombogenic risk and subsequently absorbed by the body. While beneficial for vascular closure, this mechanism can sometimes lead to complications.[8] The 6 F Angio-Seal is typically used to close arteriotomies up to 6 F (60% of our cohort), while the 8 F Angio-Seal is used for arteriotomies up to 8 F (30% of our cohort). However, some studies have evaluated the 8 F Angio-Seal as a safe and effective option for closing arteriotomies ranging from 9 to 12 F (10% of our cohort).[11]

Femoral artery occlusion may occur due to the following mechanisms: (1) deployment of the collagen plug in the intravascular compartment; (2) footplate positioning in a small vessel, bifurcation, or a highly diseased vessel insufficient to permit flow; or (3) the footplate pulling against a calcified atheroma on the posterior vessel wall, causing luminal occlusion. Small vessel diameter (<5 mm), severe atherosclerotic disease near the access site, lumen narrowing (>40%), and calcium at the posterior wall were identified as risk factors for vessel occlusion with Angio-Seal and should be considered carefully.[12,13] In our cohort, 35 patients (88%) had prior CVD, with 16 (40%) showing clinical evidence of PAD, suggesting that this condition could have contributed to the occlusion. Although some studies have shown anchor/collagen plug dysfunction as the cause without previous vascular disease.[14–16] In our cohort, symptoms mainly occurred in the acute and subacute periods after the index procedure, with claudication being the most common complaint, followed by acute limb ischemia. Methods to address Angio-Seal occlusion have varied and remain the subject of debate. Early interventions focused on surgical approaches to remove the occlusion with open atherectomy and vessel repair.[17–19] However, many patients are not candidates for surgical operations, and even still, doing so can increase the complication risk of re-intervention, with an endarterectomy-associated wound complication rate of around 8% and an overall reoperation rate of 10%.[20]

The primary interventional approach for definitive therapy was anterograde (100% of the cases) through contralateral access. The initial major hurdle in this procedure involved traversing the lesion, which was frequently completely obstructed by the collagen plug or footplate. When encountering difficulties in wire crossing, the wire escalation technique proved beneficial in facilitating anterograde wiring. In our study, a transition to retrograde wiring was necessary in 3 cases only (8%). Once the occlusion was crossed with a guidewire, atherectomy was performed in most of the procedures (83%) to debulk the collagen plug, followed by balloon dilatation.

HawkOne or TurboHawk atherectomy devices were used in 30 instances, constituting 75% of cases. The distinctive design of the atherectomy device was the determining factor for its selection, primarily because of two features: (1) the cutting unit's orientation can be adjusted to different planes by rotating the handle and (2) the nosecone reservoir effectively gathers the cut debris at the distal part. This allows for the safe removal of plug parts from the vessel, which is the predominant mechanism for vessel occlusion with Angio-Seal VCD. Moreover, the combination of debulking with directional atherectomy and balloon angioplasty was used to reestablish luminal patency and minimalize stenosis. One alternative approach was described by Dong et al., who described successful balloon dilation with thrombolysis in 29% of their cohort of 32 patients rather than debulking atherectomy of the occluded area.[21] Additionally, one case report utilized aspiration thrombectomy with full resolution of patient symptoms.[22] Distal embolization remains a potential occurrence during this endovascular intervention, and accordingly, our cohort deployed an embolic distal protection device in 19 cases (48%). Our cohort showed residual stenosis greater than 30% in 2 cases (7%). Due to the biodegradable nature of the Angio-Seal device, most cases were left without further intervention, anticipating full absorption within a few weeks. Stents were implanted in 7 cases (18%) to allow adequate lumen or to trap the embolized collagen plug, but we attempted to limit the use of stents in this anatomical area because of the concern for stent fracture from external compression in a high flexion zone area. This low usage of intravascular stenting as definitive therapy is similar to reports from other investigations.[21] Cutting balloons were effectively used in 5 cases, but their utilization may be more suited toward managing suture-based, rather than collagen-based, VCD-mediated occlusions.[23]

Our study is limited by its retrospective nature and small sample size in a single center, which may limit the generalizability of our results owing to potential selection bias. The study was not comparative because there was no control group representing the surgical approach. However, surgical exploration may be associated with increased morbidity and prolonged hospitalization. Finally, anatomical imaging (e.g., angiography or computed tomography) was not available for most patients at follow-up, which may have caused underestimation of restenosis.

5 | CONCLUSIONS

In the management of Angi-Seal VCD-related femoral artery occlusion, adjunctive use of directional atherectomy followed by balloon angioplasty was effective and safe allowing re-establishment of flow to the distal vessel with excellent immediate and long-term outcomes.

ACKNOWLEDGMENTS

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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How to cite this article: Haberman D, Czulada E, Lupu L, et al. Percutaneous Endovascular Management of Angio-Seal Related Vascular Occlusion. *Catheter Cardiovasc Interv*. 2024;104:1461-1468. doi:10.1002/ccd.31272